

Xixiao Pan

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EDUCATION

- **University of Michigan, Ann Arbor** GPA: 3.87/4.00 Sep. 2023 - May. 2025
Bachelor of Science in Engineering, Data Science Engineering, College of Engineering Michigan, USA
- **UM-SJTU Joint Institute, Shanghai Jiao Tong University** GPA: 3.62/4.00 Sep. 2021 - May 2025
Bachelor of Science in Engineering, Electronic and Computer Engineering Shanghai, China

SKILLS

- **Programming Languages:** C++, C#, JavaScript, Python, Unity, UE5, R, Matlab, SQL
- **Frameworks & Tools:** PyTorch, React, Flask, Github, Agile, AWS
- **Relevant Courses:** Computer Network, Web System, Distributed System, Computer Vision

EXPERIENCE

- **Software Development Engineer Intern, AMD**  May. 2024 - Apr. 2025 (Expected)
AI Developer Shanghai, China
 - Optimized LLM models 2x faster from FP16 to INT4 through AWQ quantization and Stable Diffusion 3 model 3x faster by fusing MultiHeadAttention in transformer on AMD GPU.
 - Created an interactive AI-driven chatbot website leveraging React for user interface for AMDers' uses.
 - Developed AI models ONNX export pipelines to deploy big models on local Windows platforms, including Llama3, Whisper, Stable Diffusion 3, and embedded models into Amuse-AI AMD software.
- **Research Intern, University of Michigan, UMTRI**  Jan. 2024 - Present
Advised by Paul Green Ann Arbor, Michigan, USA
 - Engineered a driving simulator in Unreal Engine 4 with visual occlusions to measure road safety.
 - Optimized rendering speed 4x faster with macro variation and LODs with foliage and road signs in the simulator environment with random distribution.
 - Exported realistic data of 10 km long I-94 highway in Detroit to RoadRunner for road construction.
- **AI Researcher, Shanghai Jiao Tong University, Computer Science & Engineering**  Apr. 2022 - Apr. 2023
Advised by Qinya Li Ann Arbor, Michigan, USA
 - Implemented deep learning models **ResNet-50** and **VGG** with **Pytorch**.
 - Read papers and reproduced self-supervised models **SimCLR** and **MoCo** for image assessment.

PROJECTS

- **Lipstick Expert: Web Search Engine** - *Python, NLP*  Feb 2024 - Apr. 2024
<https://lipstickexpert.netlify.app/>
 - Built a web search engine for selecting the most suitable lipstick based on price, benefits, and color. Collected dataset from YouTube and Sephora using BeautifulSoup, YouTube API, and Sephora API.
 - Achieved more than 80% accuracy by optimizing model performance using Kendall-tau distance and employing machine learning models (Naive Bayes) for sentiment analysis.
 - Employed information retrieval techniques text analysis and link analysis to calculate scores.
- **Instagram Simulation: Full-stack Web Development** - *Python, React, Flask, SQL*  Jan. 2024 - Apr. 2024
 - Created a platform with features including login and logout, posts, comments, likes, and followers.
 - Utilized SQL for database management and REST API to facilitate communication with the server.
 - Leveraged Flask and React to construct both client-side and server-side dynamic pages.
- **Virtual Reality Application in Education: DinoSnap** - *Unreal Engine 5, Agile*  Feb 2024 - Apr. 2024
<https://eecs498-group6.netlify.app/>
 - Used Unreal Engine 5 to teach players facts about dinosaurs through interacting features including finding food, riding on the dinosaur, and taking photos with the virtual camera.
 - Utilized affordance systems, raycasting, pawn possessions and collision systems for entertainment.

AWARDS & LEADERSHIP EXPERIENCE

- **Shanghai Jiao Tong University Undergraduate Excellent Scholarship** 2021 - 2022
- **Shanghai Orientation Leaders of Pre-Departure Orientation Events** - *University of Michigan* June 2024
- **Teaching Assistant** - *Shanghai Jiao Tong University* Feb. 2023 - July 2023
- **President of Nanyang Dance Association** - *Shanghai Jiao Tong University* Sep. 2022 - Aug. 2023